BEU CONNECT

Bihar Engineering University, Patna

B.Tech 7th Semester Examination, 2025

Code: 100708 ***Time: 03 Hours***

Course: B.Tech ***Subject: Biology for Engineers*** ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct option / answer the following (Any seven question only): **[2 x 7 = 14]**

(a) Which major factor differentiates DNA from RNA?
- (i) Guanine
- (ii) Cytosine
- (iii) Thymine
- (iv) Uracil

(b) Branch of science dealing with classification of organisms is called as ____
- (i) Bioautonomy
- (ii) Taxonomy
- (iii) Nomenclature
- (iv) Specialization

(c) What is the nature of charges that amino acid side chains carry?
- (i) Positive
- (ii) Negative
- (iii) Neutral
- (iv) A and B

(d) Which of the following is not an exclusive characteristic of living organisms?
- (i) Consciousness
- (ii) Movement
- (iii) Metabolism
- (iv) Reproduction

(e) The protein may consist of ____
- (i) Polar residues
- (ii) Non polar residues
- (iii) Slightly polar residues
- (iv) Polar residues and non polar residues

(f) Who is known as the "Father of Genetics"?
- (i) Morgan
- (ii) Mendel
- (iii) Watson
- (iv) Bateson

(g) Which part of the body is the control center for the nervous system?
- (i) Leg
- (ii) Stomach
- (iii) Brain
- (iv) Heart

(h) What is the function of smooth ER (SER)?
- (i) Protein production
- (ii) Provides smoothness to the cell
- (iii) Provides attachment with mitochondria
- (iv) Synthesis and transport of lipids

(i) The simplest amino acid is
- (i) glycine
- (ii) alanine
- (iii) asparagine
- (iv) tyrosine

(j) What is the nature of charges that amino acid side chains carry?
- (i) Positive
- (ii) Negative
- (iii) Neutral
- (iv) A and B

Q.2 Solve both questions : **[14]**

(a) Compare Prokaryotic and Eukaryotic cell with suitable diagram.

BEU CONNECT
LEARN • GROW • SUCCEED
Your Engineering Journey, Simplified
PYQs • SYLLABUS • RESULTS
AI DOUBT SOLVER • ATTENDANCE • GATE MOCK TESTS
Free • Fast • Made by students
GET IT ON Google Play
Scan QR to download
Search BEU Connect on Play Store

(b) How autotrophs utilize carbon and energy.

Q.3 Solve both questions : **[14]**

(a) What are enzymes and explain its role in plants?

(b) What are carbohydrates? Write any four functions of carbohydrates.

Q.4 Solve both questions : **[14]**

(a) Write short notes on fundamental differences between science and engineering.

(b) Compare Eye and Camera.

Q.5 Solve both questions : **[14]**

(a) What is Mitotic Cell division? Explain Mitosis with neat diagram.

(b) Give brief account on Dominant Epistasis with suitable example.

Q.6 Solve both questions : **[14]**

(a) Describe glycolysis in detail with suitable diagram.

(b) Explain identification and classification of microorganisms.

Q.7 Solve both questions : **[14]**

(a) What is the concept of single celled organisms?

(b) Explain briefly the sterilization and media compositions.

Q.8 Solve both questions : **[14]**

(a) Compare the exothermic and endothermic reactions.

(b) Discuss "Proteins as enzymes, transporters and structural elements".

Q.9 Write Short notes on any two of the following: **[7 x 2 = 14]**

(a) Glycolysis

(b) Working of Human Eye

(c) DNA and RNA

(d) Photosynthesis

BEU CONNECT
LEARN • GROW • SUCCEED
Your Engineering Journey, Simplified
PYQs • SYLLABUS • RESULTS
AI DOUBT SOLVER • ATTENDANCE • GATE MOCK TESTS
Free • Fast • Made by students
GET IT ON Google Play
Scan QR to download
Search BEU Connect on Play Store